LAB ASSIGNMENT - 02

- **Problem Statement**: Write a program in Java showing hierarchical inheritance with base class as Employee & derived class as Full Time Employee & Intern Employee with methods Display Salary in base class & Calculate Salary in derived classes. Calculate salary method will calculate as per increment given to full time & intern employees. Full time employee - 50% hike, Inter-employee - 25% hike. Display salary before & after hike.

- **Objective**: i) To study inheritance in Java.
ii) To study why to use inheritance.
iii) To study types of inheritance.

- **Theory**:

i) What is inheritance in Java?

→ Inheritance is an object-oriented programming mechanism where one class acquires the properties & methods of another class. The class that inherits is called a sub-class & the ~~child~~ class being inherited from is called super/parent class.

ii) Why to use inheritance?

→ Following are its uses:-

i) It promotes code reusability.
ii) Reduces code duplication.
iii) It supports method overriding
iv) It represents real-world relationships.

iii) Types in inheritance :-

i) Single inheritance: A single child class inherits properties & methods from a parent class.

ii) Multilevel inheritance: A class is derived from another derived class, forming chain of inheritance.

iii) Hierarchical inheritance: Multiple child classes inherit from the same parent class.

iv) Multiple inheritance : A class inherits behaviour from more than one interface.

v) Hybrid: A combination of 2 or more types of inheritence achieved using interfaces in Java.

• Algorithm:

1. Start.
2. Create a base class Employee with attributes salary & method display Salary().
3. Create derived class Full Time Employee that extends Employee.
4. Create derived class Intern Employee that extends Employee.
5. Implement calculate Salary() in both derived classes.
6. Apply salary hike:
   - Full time : 50%
   - Intern : 25%
7. Display salary before & after hike.
8. Stop.

• Conclusion: Thus, we have successfully implemented usage of inheritance in Java.

• FAQs:-

Q.1 Is multiple inheritence in Java? How is it achieved?

→ No, Java doesn't support multiple inheritence using classes to avoid ambiguity. However, it is achieved using interfaces, where class can implement multiple interfaces.

Q.2 What is an Is-A relationship in Java?

→ An Is-A relationship represents inheritence in Java, where a subclass is a specialized form of its superclass. It indicates that the child class inherits properties & behaviours of the parent class & can be treated as an object of the parent types.

Eg → FullTimeEmployee is an Employee.

Q.3 Are constructors & instance initialization blocks inherited by subclass?

→ No, constructors & instance initialization blocks are not inherited by subclasses in Java. This is because constructors are meant to initialize objects of their own class.

However, the constructor of the superclass can be ~~[illegible]~~ invoked from the subclass using the 'super()' keyword.

EDUCATIONAL USE